

**CHINMAYA INTERNATIONAL RESIDENTIAL SCHOOL
COIMBATORE**

SAMPLE PAPER – 2010

CLASS : 11 CBSE SCIENCE

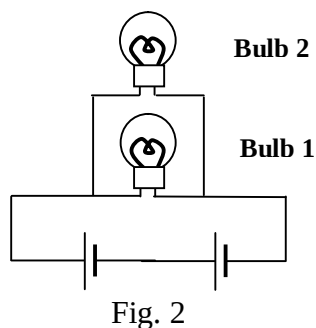
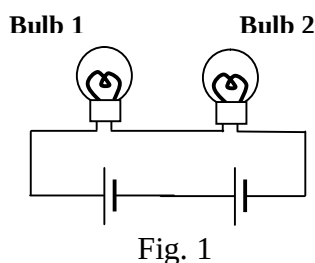
SUBJECT: PHYSICS

01. A radioactive element 'P' emits an alpha particle, and then emits a beta particle along with gamma rays. The element 'Q' thus obtained will be

- A. ${}^{Z-4}_{A-1}\text{Q}$
- B. ${}^{Z+4}_{A-1}\text{Q}$
- C. ${}^{Z-4}_{A+1}\text{Q}$
- D. ${}^{Z+4}_{A+1}\text{Q}$

ANSWER	
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02. In the circuit diagram shown below, both the bulbs are similar and have the same rating. What will happen if the circuit is rewired as shown in Fig. 2?



- A. There will be no change in the brightness of the bulbs
- B. Bulb 1 and 2 will glow dimmer
- C. Bulb 1 and 2 will glow brighter
- D. Bulb 2 will not glow at all

ANSWER	
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03. A battery is connected first across one bulb and then a second bulb is connected in series with the first. If both the bulbs are identical having the same power rating, then the brightness of the first bulb

- A. increases
- B. decreases
- C. remains the same
- D. indicates insufficient information

ANSWER	
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04. An isotope of Iodine, ${}^{131}_{53}\text{I}$ is given. What are its number of protons and neutrons respectively?

- A. 53 and 78
- B. 78 and 53
- C. 53 and 131
- D. 131 and 78

ANSWER	
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05. A stone is thrown straight upwards with a speed of 10 m/s. When it comes back to its initial height, what will be the magnitude of the stone's velocity and acceleration?

- A. 0 m/s and 0 m/s² respectively
- B. 10 m/s and 0 m/s² respectively
- C. 0 m/s and 10 m/s² respectively
- D. 10 m/s and 10 m/s² respectively

ANSWER	
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06. If a light ray is incident on a glass block along the normal to the surface, what will be its angle of refraction?

- A. can't say: data is insufficient
- B. 0°
- C. 90°
- D. 180°

ANSWER	
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